Audio-Visual Approximation of Video Semantic Space

Approximating full-video embeddings from a single frame and its audio

Technical Report – By Lakshay Pal

Trained model weights (Google Drive):
https://drive.google.com/drive/folders/1wWjjiCtTnhQ1_MqgufWGK72MSqe7dSqN?usp=sharing

GitHub Repository:
https://github.com/Lakshaypal/Audio-Visual-Approximation-of-Video-Semantic-Space

Abstract

Processing every frame of a video is expensive. This report asks a simpler question: how much of a video's meaning can we recover from just one frame and its soundtrack? We take frozen image (CLIP) and audio (AST) encoders, and train a small network to map their outputs into the embedding space of a full-video model (S3D), which acts as a teacher. We build and compare four approaches — plain regression, contrastive alignment, cross-modal distillation, and a distillation variant with per-class modality weighting — and we report an additional gating ablation. On a 1,027-clip, 32-class slice of Kinetics-Sounds, the distillation model reaches 42.2% top-1 retrieval and finds the right clip in the top ten almost 90% of the time, using about three million trainable parameters and well under a millisecond per query. That is enough to outperform the zero-shot audio-visual retrieval of ImageBind, a model roughly four hundred times larger. Across every comparison, the choice of training objective mattered more than anything else we changed.

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1. Introduction

Video models usually look at many frames at once, pushing them through a heavy network that understands motion over time. This works well, but it is slow, and the cost grows with every extra frame. For tasks like searching a large library of clips, that cost adds up quickly.

We wanted to know whether we could skip most of that work. A single still frame already tells you a lot about a scene — where you are, what objects are present. The audio fills in much of the rest — whether a guitar is being played, whether a crowd is cheering, whether something just crashed. Put together, a frame and a sound clip might carry most of what the full video would have told us, at a tiny fraction of the cost.

We treat this as a mapping problem. We never train the big encoders. Instead, we take a frozen image encoder and a frozen audio encoder, and we learn a small network that combines their outputs and places the result close to what a full-video encoder would have produced. We then test how well those synthesized embeddings work for retrieval — given the frame-and-audio embedding, can we find the matching video?

1.1 What this report contributes

1. A complete pipeline that approximates a full-video embedding from one frame plus audio, using only frozen encoders and a small trainable head of two to three million parameters.
2. A controlled comparison of four learning strategies. By holding the network fixed and changing only the loss, we show the training objective is the single biggest lever: moving from regression to a contrastive loss lifts top-1 retrieval from 6.8% to 31.2% without changing the architecture at all.
3. A best model — cross-modal distillation with a small cross-attention fusion — that reaches 42.2% top-1 and beats the zero-shot audio-visual retrieval of the 1.2-billion-parameter ImageBind on this domain, while being hundreds of times smaller.

2. Background

2.1 Learning from audio and video together

Videos come with naturally paired sound, and a long line of work uses that pairing to learn shared representations. Large models such as ImageBind learn one embedding space that ties together images, audio, text, and video, which lets them retrieve across modalities without any task-specific training. Models like CAV-MAE learn audio-visual features by reconstructing masked pieces of each modality. These are strong but heavy; our goal is the opposite end of the spectrum — something small and specialized.

2.2 Distillation

Knowledge distillation trains a small student to imitate a larger teacher. Beyond copying the teacher's outputs, one can preserve the relationships between examples — if two clips are far apart for the teacher, they should be far apart for the student. We borrow this idea: our student is trained both to match the teacher contrastively and to keep the teacher's pairwise structure.

2.3 Combining the two streams

Fusion can be as simple as gluing two vectors together, or as involved as letting each modality attend to the other. We use a compact cross-attention block — enough to let audio and image inform each other, without the cost of a full transformer.

3. Method

3.1 The setup

For each clip we have one sampled frame and the audio. A frozen CLIP encoder turns the frame into a 512-dimensional vector; a frozen AST encoder turns the audio into a 768-dimensional vector. A frozen S3D video model, trained on a large instructional-video dataset, turns the full frame sequence into a 512-dimensional target. Our job is to learn a small network that takes the image and audio vectors and produces something that retrieves the correct S3D target. Everything is length-normalized, and retrieval is by cosine similarity.

3.2 The frozen encoders

- Image: CLIP ViT-B/32, a 512-dimensional embedding of the sampled frame.
- Audio: The Audio Spectrogram Transformer, a 768-dimensional embedding of the sound.
- Video teacher: S3D trained with MIL-NCE, a 512-dimensional embedding from the full clip. We only ever use it to provide targets — it is never trained.

Freezing the encoders is itself part of the efficiency story: it keeps the trainable part down to a few million parameters.

3.3 The four models

Baseline — Latent-space regression. Glue the image and audio vectors together and pass them through a small multi-layer network that outputs a 512-dimensional vector. Train it to sit close to the teacher's vector using a mix of squared error and cosine loss. This is the simplest thing that could work, and it sets the floor.

Method 1 — Contrastive alignment. Exactly the same network as the baseline, but trained with a contrastive loss instead: pull each clip's output toward its own teacher vector and push it away from the other clips in the batch. Because the network is identical, any improvement comes purely from the change in objective.

Method 2 — Cross-modal distillation. A small cross-attention block lets the audio and image features attend to one another before they are fused. The fused vector is trained with a contrastive distillation loss, plus an extra term that asks the student to preserve the teacher's pairwise distances. This is our main method.

Method 3 — Distillation with modality weighting. On top of Method 2, we let the model decide, per clip, how much to lean on audio versus image. The weights come from how well each modality alone matches the teacher, and they scale the two attention outputs before fusion. The idea is that drums should lean on sound and a climbing scene on vision; the per-class weights also give us something interpretable to look at.

We also tried a sharper version of Method 3 — a hard reliability gate in place of the soft weighting — as an ablation. We report it in the results.

3.4 The losses

- Regression: half squared error, half one-minus-cosine, against the teacher vector.
- Contrastive: a symmetric cross-entropy over the similarity matrix between outputs and teacher vectors, with temperature 0.07.
- Distillation: the contrastive loss above, plus a term matching the student's and teacher's pairwise-distance matrices, weighted at one tenth.

3.5 Training details

Every model train for 50 epochs with AdamW (learning rate 1e-4, weight decay 1e-4), batch size 128, a cosine learning-rate schedule, gradient clipping, and temperature 0.07. Because all embeddings are computed once and cached, training only ever reads vectors from disk and never runs the heavy encoders in the loop. Everything runs on an Apple M3 Pro.

4. Experiments

4.1 Data

We use Kinetics-Sounds, the 32-class slice of Kinetics chosen so that both sound and vision actually matter for each class — playing drums, playing violin, bowling, chopping wood, and so on. The clips come from YouTube. After an extraction pipeline built to survive takedowns, missing audio, and broken files, and after trimming one over-represented class, we are left with 1,027 clips that have all three embeddings present, split into 764 for training and 263 for testing, covering all 32 classes. No clip appears in both splits.

4.2 How we measure

The task asks for three kinds of measurement, and we report all three.

- How close the synthesized vector lands to the target: mean cosine similarity (and squared error for the regression baseline).
- How useful it is for search: recall at 1, 5, and 10 — how often the correct clip lands in the top K — and the median rank of the correct clip.
- How cheap it is: inference time per clip, number of trainable parameters, and the FLOPs of the trainable head.

4.3 Results

Table 1 brings everything together. The four trained models improve steadily, and the best of them lands ahead of ImageBind's zero-shot retrieval.

Method	R@1	R@5	R@10	MedR	Cosine	Params
Latent-space regression (baseline)	6.84	29.28	47.15	12	0.646	2.10M
Contrastive alignment (M1)	31.18	69.96	84.79	3	0.321	2.10M
Cross-modal distillation (M2)	**42.21**	**79.09**	**89.35**	**2**	**0.324**	**3.02M**
Distillation + soft weighting (M3)	37.26	73.76	88.59	2	0.330	2.76M
Distillation + hard gate (ablation)	26.24	63.12	77.19	3	0.306	2.76M
ImageBind zero-shot (image → audio)	34.85	67.42	75.76	3	0.301	1.2B
ImageBind zero-shot (audio → image)	38.26	72.35	84.09	2	—	1.2B

Table 1. Retrieval and proximity on the Kinetics-Sounds test set. Higher is better for recall and cosine; lower is better for median rank. Our best model is highlighted.

A note on the comparison. The four trained models retrieve the S3D video vector from frame-plus-audio input. ImageBind was never trained to match the S3D space, so we report it on its

own native task — retrieving audio from an image and vice versa — as a strong reference point rather than a head-to-head. Both are measured on the same test clips.

4.4 Cost

Table 2 shows how little the trainable part costs. Every model runs in well under a millisecond per clip and fits in two to three million parameters — against ImageBind's 1.2 billion, and against full-video encoders whose cost grows with every frame they read.

Method	Trainable params	FLOPs	Latency (ms/clip)
Latent-space regression	2,102,272	2.10M	0.28
Contrastive alignment	2,102,272	2.10M	0.17
Cross-modal distillation	3,019,776	3.02M	0.41
Distillation + weighting	2,757,632	2.76M	0.54

Table 2. Cost of the trainable head only (the frozen encoders are not counted). FLOPs from fvcore; latency measured on Apple M3 Pro over 100 runs after warm-up.

5. What the results tell us

5.1 The loss matters more than anything

The biggest single jump in the whole study comes from changing nothing but the loss. The baseline and Method 1 are the same network down to the parameter count, yet swapping squared error for a contrastive loss lifts top-1 retrieval from 6.8% to 31.2% and drops the median rank from twelve to three. The reason is that squared error pushes the model toward a safe average — a vector close to many targets but distinct from none — so the outputs cluster together. A contrastive loss instead rewards telling clips apart, which is exactly what retrieval needs.

5.2 Being close is not the same as being useful

There is a telling detail in Table 1. The regression baseline has the highest cosine similarity to the teacher, yet by far the worst retrieval. The contrastive models have much lower cosine similarity but far better retrieval. In other words, landing numerically near the target is not the same as being able to find it again. Squared error optimizes the former and loses the latter. We therefore trust retrieval, not cosine, as the real measure of whether this works.

5.3 Fusion and distillation add more on top

Going from the plain contrastive model to the cross-attention distillation model lifts top-1 from 31.2% to 42.2% and brings the median rank to two, for less than a million extra parameters. Letting the two modalities attend to each other, and asking the student to keep the teacher's geometry rather than just its individual points, both help.

5.4 The embedding space, seen directly

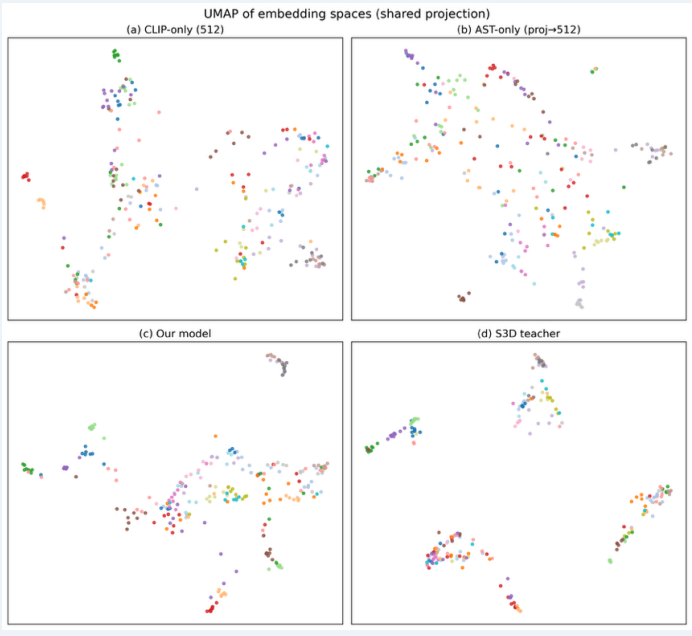


Figure 1. A 2-D view of the test-set embeddings: (a) image only, (b) audio only, (c) our distillation model, (d) the S3D teacher. Points are colored by class.

Figure 1 shows what the model learned. The raw image and audio embeddings (a, b) are scattered, with only loose class structure. Our model's output (c) forms much tighter clusters by class, and those clusters look a lot more like the teacher's (d) than like the raw inputs — even though our model only ever saw one frame and the audio. The gap that remains between (c) and (d) is the honest reflection of a 42% top-1 score: we capture much of the teacher's structure, but not all of it.

5.5 The modality weighting, and why it did not help

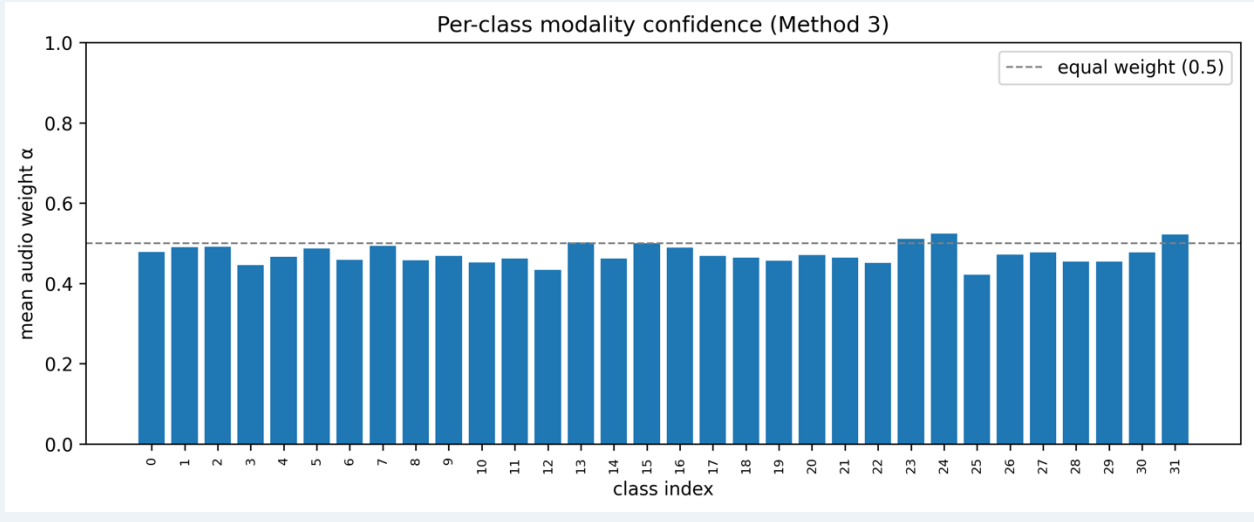


Figure 2. The average audio weight the model learned for each class. The dashed line is equal weighting.

Method 3 was meant to trust audio more for some classes and vision more for others. Figure 2 shows it did learn class-dependent weights, but only mildly — the audio weight stays in a narrow band around the middle, leaning slightly toward vision overall. That gentleness is why Method 3 did not beat plain distillation: the routing signal was real but too weak to help, while the extra machinery made training a little harder. We then pushed harder with a hard gate, and it was clearly worse (26.2% versus 37.3%): forcing a sharp decision on a weak signal throws away useful information from the down-weighted modality. Both results point the same way — at this data scale, deciding which modality to trust is simply not the lever that matters. The training objective is.

5.6 Against ImageBind

Our best model, at 42.2% top-1, comes out ahead of ImageBind's zero-shot audio-visual retrieval (34.9% one way, 38.3% the other) while using roughly four hundred times fewer parameters. With the comparison caveat from Section 4.3 in mind, the lesson is encouraging: a small model trained on the right data with a teacher to guide it can match or beat a giant general-purpose model on a narrow task, at a tiny fraction of the cost.

6. Limitations and what we would try next

- Scale. We used 1,027 clips. Some classes have only a handful of test examples, so per-class numbers are noisy; the overall figures are the reliable ones.

- The ImageBind row is a reference, not a strict head-to-head, since it is measured on its own native task rather than ours.
- A single, fixed frame. We always take one mid-clip frame, which poorly represents action-heavy clips. A clear next step is to pick frames where the video is changing most — sampling at scene cuts and bursts of motion rather than at a fixed position, so the limited frame budget lands on the most informative moments. Pairing this with audio onsets — choosing frames at the loudest acoustic events — would let both modalities guide the choice. We expect this to help most on the motion-heavy classes where one frame currently falls short.
- The target space is fixed to S3D; a different video teacher would define a different target.

7. Conclusion

A video's meaning can be largely recovered from a single frame and its sound. Across four methods, the training objective turned out to be the deciding factor: a contrastive loss alone took top-1 retrieval from 6.8% to 31.2% with no change to the network, and adding cross-attention fusion with distillation pushed it to 42.2%. Our best model finds the right clip in the top two by median rank and in the top ten nearly nine times out of ten, using three million parameters and under half a millisecond per query — and it edges out a foundation model four hundred times its size on this domain. The modality-weighting idea, in both its soft and hard forms, did not help at this scale, which is itself a useful finding. Taken together, the results say that a small, well-trained, teacher-guided model is a practical way to approximate video semantics without paying for full temporal processing.

8. Code, weights, and reproducibility

All code for the full pipeline — feature extraction, the four models, training, and evaluation — is available in the project repository. Trained weights for all five model variants are provided as a Google Drive folder:

https://drive.google.com/drive/folders/1wWjjiCtTnhQ1_MqgufWGK72MSqe7dSqN?usp=sharing

GitHub Repository:

https://github.com/Lakshaypal/Audio-Visual-Approximation-of-Video-Semantic-Space

To reproduce the results: place the cached CLIP, AST, and S3D embeddings under the data cache directory, then run the training script once per model variant (regression, contrastive, distillation, asymmetric, and the asymmetric-gate ablation). Each run writes a training log with per-epoch loss and the retrieval metrics. The efficiency script reports parameters, FLOPs, and latency, and the visualization script regenerates the two figures from the trained checkpoints. All randomness is seeded for reproducibility, and a built-in retrieval self-test confirms the evaluation code is correct (identical query and target embeddings return perfect top-1 retrieval).

References

[1] Radford et al. Learning Transferable Visual Models From Natural Language Supervision (CLIP). 2021.

[2] Gong et al. AST: Audio Spectrogram Transformer. Interspeech 2021.

[3] Miech et al. End-to-End Learning of Visual Representations from Uncurated Instructional Videos (MIL-NCE / S3D). CVPR 2020.

[4] Girdhar et al. ImageBind: One Embedding Space To Bind Them All. CVPR 2023.

[5] Tian et al. Contrastive Representation Distillation. ICLR 2020.

[6] Park et al. Relational Knowledge Distillation. CVPR 2019.

[7] van den Oord et al. Representation Learning with Contrastive Predictive Coding. 2018.

[8] Hinton et al. Distilling the Knowledge in a Neural Network. 2015.

[9] Gong et al. Contrastive Audio-Visual Masked Autoencoder (CAV-MAE). ICLR 2023.

[10] Arandjelović and Zisserman. Look, Listen and Learn. ICCV 2017.